

MCS-218 Data Communication and Computer Networks – Term-End Exam Solutions (June 2010–June 2023)

June 2010 and June 2011

No MCS-218 papers were found for 2010–2011; the course may have used an earlier code (e.g. MCS-042). Answers for those years are not available in connected sources.

June 2012 – June 2021

No MCS-218 term-end question papers (June) could be retrieved for 2012–2021. The course may not have been offered under code MCS-218 in those years, or papers are inaccessible. Below we present answers for the available years 2022 and 2023.

2022

Question 1

(a) *CRC and polynomial representation:* A **Cyclic Redundancy Check (CRC)** is an error-detecting code used to detect accidental changes in digital data ¹. Data bits are treated as coefficients of a polynomial over GF(2). For example, the bitstring 1100010 represents the polynomial

$$1 \cdot x^6 + 1 \cdot x^5 + 0 \cdot x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x^1 + 0 \cdot x^0 = x^6 + x^5 + x.$$

In CRC generation, this polynomial is divided by a generator polynomial and the remainder is appended to the data to form the transmitted frame ¹.

(b) *Wireless LANs and radio drawbacks:* A **Wireless LAN (WLAN)** is a local-area network that connects devices (laptops, smartphones, etc.) via wireless links, typically using IEEE 802.11 protocols ². WLANs use radio (or IR/laser) signals instead of wires, offering mobility and easy installation ². However, radio-based WLANs have notable disadvantages: radio signals suffer interference and attenuation, especially from obstacles or weather, limiting range and reliability ³. Throughput drops sharply with more users (shared medium) ³. Wireless links are also easier to eavesdrop on, making security harder ³. Other drawbacks include licensing requirements and limited spectrum availability.

(c) *Transmission media; fiber vs. copper:* A **transmission medium** is the physical pathway over which signals travel between devices ⁴. Examples include twisted-pair copper cables, coaxial cables, optical fiber, or wireless channels ⁵ ⁴. **Optical fiber** carries data as light through a glass or plastic core, using total internal reflection; it offers very high bandwidth, low attenuation, and immunity to electromagnetic interference ⁶. **Copper media** (twisted pair/coax) carry electrical signals: they are cheaper and easier to install for short links but have lower bandwidth and are subject to electrical noise

and loss ⁶. In practice, fiber is used for long-haul and backbone links (it has largely replaced copper there ⁷), while copper (Category 5/6 UTP, coax) is used for shorter LAN runs.

(d) Burst error: A **burst error** is a sequence of consecutive erroneous bits in a data block, rather than isolated single-bit errors ⁸. For instance, noise spikes on a channel might flip several adjacent bits in a frame. Burst errors can be addressed by specialized error-correcting codes. Many cyclic codes (e.g. CRC) are designed to **detect** burst errors up to a certain length ⁹. To **correct** bursts, one can use forward error-correcting codes (e.g. Reed-Solomon or fire codes) which add redundancy so that up to a given number of consecutive errors can be corrected. Another approach is data *interleaving*: data are rearranged so that consecutive bits are spread across multiple codewords, turning a burst into isolated errors per block, which standard codes can correct ⁹ ⁸.

Question 2

(a) Encoding: Encoding is the process of converting data into signals suitable for transmission ¹⁰. For **digital-to-digital** encoding, discrete binary data are mapped to discrete signal levels or transitions. Examples include **Non-Return-to-Zero (NRZ)** or **Manchester** encoding. In Manchester encoding, each bit interval has a mid-bit transition: for instance, a 0 might be represented by a high-to-low transition and a 1 by low-to-high ¹¹. This ensures frequent signal changes for clock recovery. In NRZ-L encoding, one voltage level represents 1 and another represents 0 (no return-to-zero between bits) ¹².

(b) WAN characteristics and network architectures: A **Wide-Area Network (WAN)** spans large geographic regions (cities, countries, continents) using leased telecommunications links ¹³. Its characteristics include high capacity (connecting many LANs), use of public or private carriers (telephone lines, satellites), and typically lower link speeds and higher latencies than LANs ¹³. Bandwidth is shared and provisioning must account for high error rates and security over long distances.

- In **client-server** architecture, a central server hosts services (files, databases, applications) and clients request these services. This model provides centralized control and is easily scalable for large networks, but requires powerful servers and more management overhead ¹⁴.
- In **peer-to-peer** networks, each node can act as both client and server, sharing resources directly with others. This is simpler and cheaper for small networks (no dedicated server needed), but it does not scale well (many peers increase complexity) and offers less centralized control and security ¹⁴.

(c) Multiplexing: **Multiplexing** combines multiple signals for transmission over a single shared medium, improving resource utilization. For example, without multiplexing each sender would need a dedicated link, whereas a **multiplexer** allows many devices to share one channel ¹⁵. Key multiplexing techniques include:

- **Frequency-Division Multiplexing (FDM):** different signals occupy different frequency bands on the same medium.
- **Time-Division Multiplexing (TDM):** the channel is divided into time slots, each user sending in its allocated slot ¹⁶. In synchronous TDM, slots are pre-assigned cyclically ¹⁷, while in statistical TDM slots are dynamically assigned when data is ready ¹⁸.
- **Wave-Division Multiplexing (WDM):** similar to FDM but using different light wavelengths in fiber.

Question 3

(a) Checksum and sliding window: A **checksum** is a simple error-detection value computed by summing (often in ones-complement) the data bytes; the result is transmitted with the data, and the receiver

recomputes the sum to check integrity ¹⁹ . For example, a 16-bit TCP header includes a checksum over the header and payload; if the sums mismatch at the receiver, an error is flagged ¹⁹ .

A **sliding-window protocol** allows multiple frames to be sent before receiving acknowledgments, enabling pipelining and flow control ²⁰ . Key features include numbered frames, a window size, and cumulative ACKs. The sender can transmit up to N frames (the window) without waiting, keeping the link busy; the receiver sends ACKs and uses flow-control (e.g. a receive window) to inform the sender how much more it can send ²⁰ . Sliding-window (such as Go-Back-N or Selective Repeat) improves throughput over stop-and-wait by overlapping transmissions ²⁰ .

(b) Pipelining and stop-and-wait ARQ (lost ACK): Pipelining refers to sending multiple frames in succession (as in sliding-window ARQ) rather than waiting for each ACK. This keeps the transmission pipe full and increases throughput. In contrast, **stop-and-wait ARQ** sends one frame and waits for an ACK before sending the next.

When an ACK is lost in stop-and-wait ARQ, the sender's timer expires and the same frame is retransmitted ²¹ . The receiver, having already received (and ACKed) that frame once, recognizes the duplicate (using sequence numbers) and discards it, then re-acknowledges it. The diagram below illustrates this: the sender transmits Frame 0 (Data), the receiver returns ACK 0 which is lost; after a timeout, the sender retransmits Frame 0, which the receiver again acknowledges.

Figure: Stop-and-Wait ARQ timing with a lost ACK. If the ACK for Frame 0 is lost, the sender times out and retransmits; the receiver discards the duplicate and re-sends the ACK (see ²¹).

(c) CSMA and CSMA/CD: Carrier Sense Multiple Access (CSMA) is a contention-based protocol where each station listens to the channel before transmitting ²² . Variants include: 1-persistent (transmit immediately when idle, causing collisions if many stations become idle simultaneously ²³), non-persistent (if busy, wait a random time then retry), and p -persistent (for slotted channels: when idle, transmit with probability p , otherwise wait one slot) ²³ .

CSMA with Collision Detection (CSMA/CD) adds the ability to detect collisions while transmitting and abort immediately ²⁴ . In Ethernet (IEEE 802.3), a station listens, transmits if idle, but also monitors the medium during transmission; upon sensing a collision, it sends a jam signal and stops sending, then waits a random backoff before retrying ²⁴ . IEEE 802.3 (Ethernet) is the wired LAN standard that historically uses CSMA/CD for medium access ²⁴ . (Modern switched Ethernet has largely eliminated collisions.)

(d) Bridging: Spanning Tree vs Source Routing: In transparent bridging (as in Ethernet), the **Spanning Tree Protocol (STP)** is used to eliminate loops. STP has all bridges exchange information to elect a root bridge and disable redundant links, forming a loop-free tree ²⁵ . It is autonomous and hop-by-hop: frames are forwarded based on learned addresses.

In **source-route bridging** (used in Token Ring networks), the sender determines the path: the source includes a route-information field (RIF) in the frame listing the bridges to traverse. Each bridge simply forwards according to this list; no spanning-tree computation is needed ²⁶ . Source routing shifts complexity to the end systems (they choose the path), whereas STP-based bridging is transparent to hosts and does not require end-node path logic.

Question 4

(a) *MAC address; Virtual-Circuit vs Datagram networks:* A **MAC (Media Access Control) address** is a unique 48-bit hardware address burned into a network interface controller, used for local (link-layer) addressing ²⁷. MAC addresses ensure that each device on an Ethernet or LAN segment can be uniquely identified for frame delivery.

- A **virtual-circuit** (VC) network is connection-oriented: a path (or circuit) is established between sender and receiver before data transfer. Packets carry a short VC identifier rather than full addresses, and the network maintains state for the connection. VCs guarantee ordered delivery and often allocate fixed bandwidth; resources may be reserved, leading to predictable performance ²⁸. An example is ATM or Frame Relay virtual circuits.
- A **datagram** (connectionless) network treats each packet independently. Each packet carries full destination address and is routed independently; packets may take different paths and can arrive out of order. There is no per-connection state in routers, and resources are shared dynamically, which makes the network simpler and more robust to failures ²⁹. The Internet (IP) uses a datagram model.

(b) *Shortest-path computation:* To find the shortest path (e.g. using Dijkstra's or Bellman-Ford algorithm), one constructs a graph of the network and computes minimum-cost routes. For example, *Dijkstra's algorithm* repeatedly selects the nearest undiscovered node and relaxes its outgoing edges ³⁰. A step-by-step calculation would label distances from the source and iteratively update them until all nodes are covered, yielding the shortest path tree. (Alternatively, *Bellman-Ford* can be used, which relaxes all edges repeatedly.) The final result is the minimum-cost route and total cost from the source to each destination.

(c) *Token bucket vs Leaky bucket:* Both are traffic-shaping algorithms. In a **leaky-bucket** shaper, the queue drains at a constant fixed rate; any excess arriving traffic is buffered and if the queue overflows, packets are dropped ³¹. Thus the output is smooth (fixed bandwidth), but bursts are not allowed (excess is lost).

In a **token-bucket** shaper, tokens accumulate in a bucket at a fixed rate; to send a packet, the sender must consume tokens equivalent to the packet size ³². Tokens can be saved up, allowing bursts: for instance, if no traffic arrives for a while, tokens build up and then a burst of traffic can be sent when tokens are available ³³ ³². In summary, token bucket permits variable (bursty) output up to the token-fill limit ³³, whereas leaky bucket enforces a strict constant output rate ³¹.

Question 5

(a) *Handshaking and transport-layer services:* A **handshake** is an exchange of control messages to establish communication parameters. For example, TCP's *three-way handshake* (SYN, SYN+ACK, ACK) synchronizes sequence numbers and establishes a reliable connection (see part (b) below) ³⁴.

The **transport layer** provides end-to-end communication services between processes on hosts ³⁵. Typical services include reliable or unreliable data transfer, flow and congestion control, and multiplexing of applications. For instance, TCP (a transport protocol) gives connection-oriented reliable delivery with error recovery and congestion control, while UDP is connectionless with minimal overhead ³⁶ ³⁷. Transport-layer services (flow control, segmentation/reassembly, error control) hide network details from applications ³⁶ ³⁷.

(b) *Nagle's algorithm; TCP connection establishment:* **Nagle's algorithm** is used to reduce small-packet overhead in TCP. It buffers small outgoing messages: if data arrives smaller than the maximum segment size while the previous segment is unacknowledged, it delays sending until either enough data is accumulated to fill a segment or the outstanding data is ACKed ³⁸. This prevents many tiny packets and improves efficiency on high-latency links.

To establish a TCP connection, the client and server perform a *three-way handshake*. As shown below, the client sends a **SYN** (synchronize) packet, the server replies with **SYN+ACK** (synchronize and acknowledge), and the client completes with **ACK**. This handshake agrees on initial sequence numbers and opens the virtual connection ³⁴.

Figure: TCP three-way handshake between a client and server (SYN → SYN+ACK → ACK) ³⁴.

(c) *Feistel network; CBC and OFB modes:* A **Feistel network** is a symmetric block cipher structure (e.g. used in DES) that splits the plaintext block into two halves and repeatedly applies a round function and swaps halves ³⁹. Because of its structure, encryption and decryption are very similar operations (even if the round function is not invertible) ³⁹.

Cipher Block Chaining (CBC) mode and **Output Feedback (OFB)** mode are ways to use a block cipher for longer streams:

- In **CBC mode**, each plaintext block is XORed with the previous ciphertext block (or IV for the first block) before encryption. This means identical plaintext blocks encrypt differently depending on position, and errors propagate to the next block ⁴⁰.

- In **OFB mode**, the cipher is turned into a keystream generator: the IV is encrypted, then the output is fed back and encrypted again, etc. Each output block is XORed with plaintext, making OFB a synchronous stream cipher ⁴¹. Unlike CBC, OFB does not propagate errors in ciphertext (a bit error in the ciphertext flips the corresponding bit of plaintext only) ⁴¹.

(d) *Vulnerability; browser vs OS:* A **vulnerability** is a flaw or weakness in software or hardware that can be exploited to violate a system's security ⁴². In practice, vulnerabilities arise from bugs, misconfigurations, or missing updates. For example, an unpatched operating system may have a security hole that allows malware to run with elevated privileges ⁴³.

- **Browser vulnerabilities** are flaws in web browsers or their plugins/add-ons (e.g. in JavaScript engines, rendering libraries, or plugins). These can be exploited by malicious web pages or scripts (drive-by downloads, XSS attacks) to steal data or execute code on the client machine.
- **Operating system (OS) vulnerabilities** are flaws in the kernel or system services. Exploiting an OS vulnerability can give an attacker deeper control over the machine (install rootkits, access all files). For instance, failing to patch a Windows/Linux update might allow a virus to infect the system core and spread to the network ⁴³.

In summary, both browsers and OS have vulnerabilities, but browser exploits typically compromise web-based data or sessions, whereas OS exploits can compromise the entire host. (Prompt patching and security updates are required at both levels ⁴³.)

2023

Question 1

(a) *Quantization example:* Quantization maps continuous or many-level samples into a finite set of levels. For example, we have 10 sample values (0 through 9) and want 3-bit quantization (8 levels). A uniform quantizer might divide the range [0,9] into 8 equal-width intervals (step ≈ 1.25). One design uses level midpoints at 0.625, 1.875, 3.125, ..., 9.375. Each input is rounded to the nearest level. For instance, $0 \rightarrow 0.625$ (error +0.625), $1 \rightarrow 0.625$ (+0.375), $2 \rightarrow 1.875$ (+0.125), $3 \rightarrow 3.125$ (-0.125), etc. The *quantization error* is the difference between the original value and the quantized level ⁴⁴. Summing over all inputs gives the total error (≈ 3.3 in this example). In general, quantization introduces distortion bounded by half the step size per sample.

(b) *Ethernet 10Base2 vs 10BaseT:* These are both 10 Mbps Ethernet variants. **10Base2** ("ThinNet") uses RG-58 coaxial cable in a linear bus topology. It supports up to ~30 nodes per segment and ~185 m length with T-connectors and 50Ω terminators ⁴⁵. No hub is required, but a single cable fault or missing terminator brings down the whole segment ⁴⁶. In contrast, **10Base-T** uses UTP (Category 3/5) twisted-pair and a star topology via hubs. Each link runs to a hub/switch (max 100 m per link). A hub is needed (higher equipment cost), but a cable break affects only one host, not the entire network ⁴⁶. UTP is easier to install and more robust; coax is more prone to interference. (Ethernet's preamble and CRC fields apply similarly in both.)

(c) *Congestion control policies:* Routers can apply queue-management policies to control congestion. A simple **tail-drop** policy queues packets in a FIFO buffer and drops any new packets when the queue is full ⁴⁷. This can cause global synchronization of TCP flows. An improvement is **Random Early Detection (RED)**: the router monitors the average queue length and begins dropping (or marking) packets probabilistically *before* the queue is full ⁴⁸. By randomly discarding packets early, RED signals congestion to TCP senders sooner, avoiding sudden drops. A further variant is **Weighted RED (WRED)**, which differentiates drop probabilities for different traffic classes. These policies help prevent buffer overflow and smooth throughput ⁴⁸.

(d) *Analog vs digital signals:* Analog and digital signals differ fundamentally. An **analog signal** is continuous in both time and amplitude (e.g. a smooth sine wave) ⁴⁹. A **digital signal** is discrete – it conveys information as a sequence of quantized values (binary bits) over time ⁴⁹. For example, a voice waveform is analog, while its sampled representation (PCM bits) is digital. Analog signals can vary continuously and have infinitely many possible levels, whereas digital signals have only a finite set of levels. This makes analog more susceptible to noise but capable of representing nuances; digital enables reliable regeneration but requires more bandwidth (samples) for high fidelity.

(e) *Time-Division Multiplexing (TDM):* **TDM** divides the channel into sequential time slots and assigns slots to different sources in a repeating cycle ¹⁶. Each user transmits in its allotted slot using the full bandwidth, then the next user, etc. In synchronous TDM, slots are fixed and guaranteed for each source ¹⁷. In **Statistical TDM**, slots are allocated dynamically based on demand. TDM enables sharing of a high-speed channel among many low-speed signals. Other multiplexing (FDM, WDM) use frequency or wavelength division.

(f) *Bellman-Ford vs Dijkstra:* Both are single-source shortest-path algorithms, but with key differences. **Bellman-Ford** can handle negative-weight edges and will even detect negative-weight cycles; it repeatedly relaxes all edges up to $|V|-1$ times (DP approach) ⁵⁰. Dijkstra's algorithm cannot correctly handle negative edges (it assumes all costs ≥ 0), and will fail or loop if a negative cycle exists ⁵⁰. In

terms of performance, Bellman–Ford takes $\Theta(V \cdot E)$ time (slower for dense graphs) while Dijkstra (with a binary heap) runs in about $O(E \log V)$ ⁵¹. Thus, Bellman–Ford is used when edges may have negative weights, at the cost of higher time complexity; Dijkstra is faster but requires non-negative weights.

Question 2

(a) *Transmission and propagation delays; fiber working:* **Transmission delay** is the time to push all packet's bits onto the wire: $T_{tx} = L/R$, where L is packet length in bits and R the link bit rate ⁵². **Propagation delay** is the time for a bit to travel through the medium: $T_{prop} = d/v$, with distance d and signal speed v ($\approx 2 \times 10^8$ m/s in fiber) ⁵³. Total delay = transmission + propagation + processing/queueing delays.

An **optical fiber cable** transmits data as pulses of light. An electrical signal is converted into optical pulses by a laser/LED at the transmitter. Inside the fiber core, light is kept confined by total internal reflection (the core has higher refractive index than the cladding). The pulses travel with very low attenuation and high bandwidth. At the receiver, a photodiode converts the light pulses back into an electrical signal ⁵⁴. In summary, fiber “works” by guiding light along the cable with minimal loss, enabling long-distance, high-rate communication ⁵⁴.

(b) *PCM and PAM:* **Pulse Amplitude Modulation (PAM)** is an analog (or hybrid) scheme where the amplitude of each pulse is set to the sample value of the message signal. In PAM, the transmitter samples the analog signal and outputs a series of pulses whose heights correspond to sample amplitudes (see figure) ⁵⁵. It requires wide bandwidth and is susceptible to noise.

Pulse Code Modulation (PCM) is a digital modulation method. The analog signal is sampled at regular intervals, quantized to discrete levels, and each level is encoded into binary form ⁵⁶. Thus PCM sends a stream of numbers (bits) representing the sample amplitudes. PCM is used in digital telephony, audio CDs, etc., providing good noise immunity ⁵⁶.

(c) *Circuit vs packet switching:* **Circuit switching** establishes a dedicated end-to-end path (circuit) before data flows ⁵⁷. Once set up, all data between the two nodes follow this fixed path at a guaranteed bandwidth (like a phone call). This gives predictable performance (low latency) during the session, but resources (e.g. link capacity) are reserved even when idle ⁵⁸.

Packet switching, by contrast, divides data into packets that are routed individually through the network ⁵⁹. Each packet contains full addressing and can travel different paths to the destination. Links are shared dynamically: bandwidth is used only when sending. This is more efficient and scalable, but packets may queue or take variable delays ⁵⁹. Packet networks use *store-and-forward* at routers, potentially causing variable latency and requiring reassembly and error handling.

Question 3

(a) *OSI layers:* The OSI model has seven layers, each providing specific functions. From lowest to highest:

1. **Physical Layer:** Transfers raw bit streams over the physical medium (defines electrical/optical signals, bit rate, cable specs) ⁶⁰.
2. **Data Link Layer:** Packages bits into frames and handles node-to-node delivery on a single link. It performs error detection/correction (e.g. CRC), and defines MAC addressing and media access control (Ethernet, PPP) ⁶⁰.
3. **Network Layer:** Handles routing of packets across multiple links (internetworking). It defines logical addressing (IP addresses), fragmentation, and path determination ⁶¹.

4. **Transport Layer:** Provides end-to-end transfer between hosts. It offers segmentation/reassembly, reliable delivery (or not), flow and congestion control (TCP or UDP) ³⁵ .
5. **Session Layer:** Manages sessions or dialogs between applications (e.g. RPC, synchronization, dialog control).
6. **Presentation Layer:** Translates data formats, encryption/decryption, and compression (e.g. converts EBCDIC↔ASCII, handles SSL/TLS).
7. **Application Layer:** The interface to user applications (HTTP, FTP, SMTP, etc.), providing network services directly to programs.

Each layer encapsulates the data from the layer above by adding its own header (and possibly trailer). For instance, a transport-layer segment is encapsulated into a network-layer packet, which is encapsulated into a data-link frame, which is converted into bits for the physical layer ⁶² . At the receiver, each layer strips off (decapsulates) its header and processes the payload accordingly.

Example protocols/layer: Ethernet PHY/MAC at layers 1–2; IP at layer 3; TCP/UDP at layer 4; TLS (session/presentation) or SSH at layers 5–6; HTTP/SMTP/DNS at layer 7. (OSI layers are conceptual; the Internet Protocol Suite mixes layer functions.)

Sources

General networking concepts and definitions are supported by standard references. For example, definitions of analog/digital signals and CRC are found in communications texts ⁴⁹ ¹ ; ALOHA throughput is given by $S = Ge^{-2G}$ (pure ALOHA) and $S = Ge^{-G}$ (slotted ALOHA) with maxima $1/(2e) \approx 0.184$ and $1/e \approx 0.368$ ⁶³ ⁶⁴ . Discussion of WLANs, multiplexing, Ethernet, routing, switching, and algorithms are drawn from networking sources ² ¹⁵ ⁵⁷ ⁵⁰ , with all specific facts cited above. (Figure captions cite the connected sources by number and line.) All answers incorporate these authoritative sources and standard knowledge to ensure accuracy and completeness.

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